

ST QUENTIN ROUPY AP, France

WMO: 070610

Lat: 49.8183N

Lon: 3.2061E

Elev: 331

StdP: 14.52

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.6	25.1	15.9	12.4	24.1	20.0	15.2	26.6	30.7	49.1	27.4	48.2	7.5	40	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	85.1	68.4	81.2	67.1	77.8	65.5	70.2	81.7	68.3	78.5	66.6	75.4	8.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	98.2	75.2	64.6	92.2	72.4	63.1	87.5	70.6	34.3	82.0	32.7	78.7	31.4	75.9	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	21.1	18.7	DB	17.3	92.4	5.9	4.2	13.0	95.4	9.6	97.9	6.3	100.2	2.1	103.2
			WB	16.5	74.0	5.7	1.9	12.5	75.4	9.2	76.5	6.0	77.5	1.8	78.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	38.1	39.5	44.5	49.6	55.7	60.8	65.0	64.5	59.1	52.5	44.6	39.3	(6)
(7)		DBStd	11.33	7.38	6.59	6.00	6.31	5.96	5.59	5.68	5.19	5.10	6.02	6.38	7.13	(7)
(8)		HDD50	1513	372	296	187	82	14	0	0	0	1	43	183	335	(8)
(9)		HDD65	5221	835	714	635	462	296	151	72	72	189	386	613	796	(9)
(10)		CDD50	1940	2	2	17	70	190	325	465	449	274	122	20	4	(10)
(11)		CDD65	172	0	0	0	0	7	25	72	56	12	0	0	0	(11)
(12)		CDH74	1701	0	0	0	11	72	270	709	541	94	4	0	0	(12)
(13)		CDH80	460	0	0	0	0	8	63	207	170	12	0	0	0	(13)
(14)	Wind	WSAvg	9.1	10.7	10.5	10.2	9.1	8.5	7.7	7.9	7.7	8.1	9.1	9.6	10.3	(14)
(15)	Precipitation	PrecAvg	28.4	2.3	2.1	2.0	1.9	2.3	2.5	2.4	2.8	2.1	2.5	2.5	2.9	(15)
(16)		PrecMax	38.4	6.3	4.4	6.7	5.0	5.0	4.8	4.8	6.7	4.5	5.4	5.2	7.4	(16)
(17)		PrecMin	21.2	0.3	0.5	0.2	0.1	0.1	1.1	0.7	0.3	0.4	0.9	0.6	1.1	(17)
(18)		PrecStd	4.3	1.2	1.1	1.3	1.1	1.3	1.2	1.0	1.5	1.1	1.1	1.1	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	57.2	66.0	75.4	80.5	86.7	90.8	91.2	81.7	72.4	61.7	56.2	(19)
(20)		MCWB	52.5	50.3	54.0	59.8	68.0	70.6	69.9	70.2	66.9	63.3	58.0	53.5	(20)	
(21)		2%	DB	53.0	53.3	61.2	70.2	75.8	80.8	85.2	85.0	76.4	67.5	58.1	53.8	(21)
(22)		MCWB	50.7	49.4	52.7	58.1	63.5	68.1	68.0	68.1	64.7	61.1	55.3	51.7	(22)	
(23)		5%	DB	50.3	51.1	57.1	64.8	71.6	76.8	81.4	80.3	71.9	63.9	55.8	51.8	(23)
(24)		MCWB	48.1	48.2	50.7	54.8	61.4	66.0	66.8	66.3	62.5	59.0	53.5	49.9	(24)	
(25)		10%	DB	48.1	49.1	54.0	60.9	67.5	72.6	77.6	75.7	68.6	61.2	53.7	49.5	(25)
(26)		MCWB	46.2	46.6	49.5	52.9	58.9	63.5	65.5	64.0	60.8	57.5	51.8	47.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	52.0	55.9	62.8	69.3	72.6	72.0	72.6	68.5	64.4	58.9	54.2	(27)
(28)		MCDWB	54.9	55.0	61.8	73.3	78.7	83.2	84.7	87.7	78.0	70.2	61.0	55.5	(28)	
(29)		2%	WB	50.9	50.1	53.9	58.6	65.1	69.1	69.9	69.6	66.0	61.8	55.7	52.0	(29)
(30)		MCDWB	52.7	52.2	59.1	67.2	73.2	77.9	81.9	81.7	73.5	66.0	57.6	53.6	(30)	
(31)		5%	WB	48.4	48.5	52.0	56.0	62.2	66.9	68.1	67.4	63.9	60.0	53.8	50.1	(31)
(32)		MCDWB	50.0	50.8	56.1	63.3	69.5	75.2	78.7	77.4	70.0	63.3	55.5	51.6	(32)	
(33)		10%	WB	46.4	46.7	49.9	53.8	59.8	64.6	66.2	65.5	62.2	58.1	51.9	48.0	(33)
(34)		MCDWB	48.0	48.8	53.4	59.5	66.1	71.6	75.2	73.5	67.4	60.7	53.6	49.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.3	10.3	13.8	17.2	17.0	18.2	19.1	18.6	16.7	13.5	9.6	8.1	(35)	
(36)		5% DB	MCDBR	9.6	12.9	19.9	23.2	23.7	24.9	25.2	26.0	21.5	16.2	10.8	9.0	(36)
(37)		MCWBR	8.6	10.0	12.8	13.1	13.3	13.4	11.0	11.5	11.4	10.4	8.7	8.2	(37)	
(38)		5% WB	MCDBR	9.4	11.7	16.6	20.4	20.9	22.6	22.0	22.7	18.5	14.7	10.1	9.0	(38)
(39)		MCWBR	8.8	9.7	11.4	12.3	12.6	13.1	10.5	11.2	10.7	10.0	8.5	8.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.343	0.377	0.406	0.415	0.414	0.414	0.404	0.377	0.361	0.333	0.318	(40)	
(41)		taud	2.413	2.375	2.269	2.203	2.216	2.246	2.245	2.297	2.359	2.407	2.417	2.381	(41)	
(42)		Ebn at Noon	228	252	260	265	268	268	266	263	259	241	220	210	(42)	
(43)		Edh at Noon	20	27	35	42	43	42	41	37	32	26	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	272	502	874	1365	1598	1761	1671	1432	1103	648	321	220	(44)	
(45)		RadStd	30	85	112	157	156	165	156	141	89	54	33	32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page